

ADITYA VERMA

Fremont, CA

(510) 944-4583 | verma231@purdue.edu | [adiverma231.github.io](https://github.com/adiverma231) | U.S. Citizen

Education

Purdue University

Bachelor of Science in Electrical Engineering, GPA 3.9/4.0

West Lafayette, IN

Aug 2024 - May 2028

- Relevant Coursework: Digital System Design Lab, Electromechanical Motion Devices, Analog IC Design, VLSI Design, Python for Data Science, Advanced C Programming

Experience

Honeywell

Embedded Engineer Intern

Northford, CT

Jun 2025 - Aug 2025

- Contributed to control panel firmware development in C++, implementing symbolic logic processing and USB data reading via Qt Creator.
- Created and followed rigorous test procedures to validate LED timings, power supply performance, and strobe light functionality.
- Conducted comprehensive circuit tests and debugged PCBs using advanced lab equipment, including oscilloscopes, digital logic analyzers, and high-voltage AC/DC power supplies.

Purdue System-on-Chip Extension Technologies

Undergraduate Researcher

Aug 2025 - present

- Designed a development board in Altium for post-silicon validation of Purdue's custom RISC-V SoC, replacing temporary test setups with a proper microcontroller-style board for 7+ chip peripherals.
- Captured full hierarchical schematic with 20+ components across 6 subcircuit sheets and no formal datasheet, reverse-engineering pin assignments for power, clock, and memory from scratch.
- Length-matched address and data buses for 2 external 500Mb SRAM ICs, achieving $< 25ns$ timing margin at 25 MHz.
- Streamlined initial board design down from 6 layers to 4, and got approval from manufacturer for fabrication.

Projects

Purdue Solar Racing

Embedded Systems Lead Engineer

Mar 2025 - present

- Led 5 engineers to design controller and steering wheel PCBs for a solar-powered race car, delivering prototypes 2 weeks early through weekly syncs and task delegation.
- Programmed STM32 microcontroller in C to convert rotary encoder and button signals to CAN messages, reducing signal latency by 40%.
- Tested 3 custom PCBs using a lab oscilloscope and logic analyzer, finding and fixing several bugs for 100% functionality.
- Oversaw production of 4 additional boards, delivering them over a month ahead of schedule.

Agrovollo Drone

Jun 2025 - Jan 2026

- Collaborated with 3 peers to build a UAV for crop field spectral analysis, helping farmers increase yields by identifying health issues early.
- Engineered custom microcontroller PCB in Altium under 10 cm² constraints.
- Debugged STM32 firmware to handle sensor data and telemetry, achieving stable flight in just 3 test runs.
- Trained PyTorch and TensorFlow computer vision models on 1500+ images to assess crop health from live video feed, with a detection accuracy of 90%.

Technical Skills

Languages: Python, C/C++, MATLAB, Verilog

Tools: Solidworks, Siemens NX, KiCAD, Altium, LTspice, UNIX, Virtuoso, UART, CAN

Patents: Ocean Plastic Scalable, Modular, Autonomous Retrieval Technology (US63/403238 Granted May 2025)

Certifications: PCEP Certified Entry-Level Python Programmer (kGJ3.RNjJ.5X7K)